



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 3 • STANDARD 6.AT.3

# Ratio Reasoning: Convert Measurements Between Systems

Lesson 3-7 • Teacher Copy

**Name:** Type your name

**Date:** Today's date

**Class:** Class period

## My Notes

Level 2 Standard



### TODAY'S OBJECTIVES

**Content Objective:** I can use ratio reasoning to convert measurements between the customary and metric systems.

**Language Objective:** I can explain a conversion between systems using the words conversion factor, approximately, customary, and metric.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Convert Measurements Between Systems

Conversion factor

Customary system

Metric system

Approximately

Ratio table



Tap any word to see what it means and a picture.

1

Rewriting a customary measurement as a metric one gives an

answer, not an exact one.

2

The ratio that trades one unit for another is a

.

3

Inches, pounds and gallons belong to the

system.

4

Centimeters, grams and liters belong to the

system, built on tens.

5

Close to but not exactly equal is

— written with  $\approx$ .

6 A table whose every column holds the same comparison is a

  .

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

EXPLAIN

**Is 100 kilometers per hour faster or slower than 100 miles per hour, and how do you know?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

☐ **Convert Measurements Between Systems** — Rewriting a customary measurement as a metric one, or the reverse, so two amounts can be compared.  
*Convertir medidas entre sistemas — Reescribir una medida usual como métrica, o al revés, para poder comparar dos cantidades.*

☐ **Conversion factor** — The ratio that links a unit in one system to a unit in the other, such as 1 mile to about 1.609 kilometers.  
*Factor de conversión — La razón que relaciona una unidad de un sistema con una del otro, como 1 milla a unos 1.609 kilómetros.*

☐ **Customary system** — The measurement system used in the United States: inches, feet, miles, ounces, pounds, cups, quarts, gallons.  
*Sistema usual — El sistema de medidas que se usa en Estados Unidos: pulgadas, pies, millas, onzas, libras, tazas, cuartos y galones.*

☐ **Metric system** — The measurement system used in most of the world, built on tens: centimeters, meters, kilometers, grams, kilograms, liters.  
*Sistema métrico — El sistema de medidas que se usa en casi todo el mundo, basado en decenas: centímetros, metros, kilómetros, gramos, kilogramos y litros.*

☐ **Approximately** — Close to, but not exactly equal. Conversions between systems are almost always approximate, so they use the  $\approx$  sign.  
*Aproximadamente — Cercano, pero no exactamente igual. Las conversiones entre sistemas casi siempre son aproximadas y usan el signo  $\approx$ .*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**The conversion factor is 1 km  $\approx$  \_\_\_\_ mi. Scaling that up, 100 kilometers  $\approx$  \_\_\_\_ miles, so the sign means about \_\_\_\_ miles per hour. That is \_\_\_\_ than 100 miles per hour, because in one hour you would travel \_\_\_\_ distance.**

*Di tu respuesta primero: Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** Divide the total cost by the number of pounds to find the cost per 1 pound.

**Evidence:** Students divide \$18 by 3 to get \$6 per pound and explain the unit rate lets them find the cost of any number of pounds.

**Reasoning:** This evidence connects the definition of convert measurements between systems to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **The conversion factor is**  $1 \text{ km} \approx \underline{\hspace{1cm}} \text{ mi}$ . **Scaling that up, 100 kilometers  $\approx$  \_\_\_\_\_ miles, so the sign means about \_\_\_\_\_ miles per hour. That is \_\_\_\_\_ than 100 miles per hour, because in one hour you would travel \_\_\_\_\_ distance.**

- **My answer is** \_\_\_\_\_.

*Mi respuesta es* \_\_\_\_\_.

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#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is** \_\_\_\_\_.

*Mi afirmación es* \_\_\_\_\_.

- **I know because** \_\_\_\_\_.

*Lo sé porque* \_\_\_\_\_.

- **So** \_\_\_\_\_.

*Entonces* \_\_\_\_\_.

## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
- **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*
- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*
- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*
- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

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## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.



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## Answer Key & Teacher Guide

1. **Notes 1:** Convert Measurements Between Systems
2. **Notes 2:** Conversion factor
3. **Notes 3:** Customary system
4. **Notes 4:** Metric system
5. **Notes 5:** Approximately
6. **Notes 6:** Ratio table
7. **Watch for this mistake:** *A common mistake converting between systems is comparing the bare numbers without converting at all — deciding that 5 kilometers and 5 miles are the same distance because both say 5. They are not: 5 kilometers  $\approx$  3 miles. A second mistake is writing = instead of  $\approx$ . Conversion factors between systems are rounded (1 mile is about 1.609344 kilometers), so the result is approximate no matter how carefully you multiply.*
8. **Try It 1:** — *Each pair is a conversion factor between the two systems, and each one is approximate.*
9. **Try It 2:** A. The conversion factors are rounded, so results are approximate — *One mile is about 1.609344 kilometers — a decimal that never ends neatly. Any usable factor is rounded, so the answer is approximate.*